Examiner reports

Q1.

In part (a)(i), many candidates were unable to produce a correctly expressed solution, despite dealing correctly with the differentiation. The main error was the omission of brackets around
$8x + 3$. Part (a)(ii) was well answered by the majority of candidates. Part (b)(i) was well answered by the majority of candidates, although a common error was $6y^2 + \frac{1}{x}$. Most candidates correctly substituted $y = 1$ and obtained 7, although 6 was a common evaluation; a substitution of $y = 2$ was also seen. Many candidates who obtained 7 then proceeded, incorrectly, to write $(y – 1) = 7(x – 2)$ although a few candidates correctly wrote "$(x – 2) = 7(y – 1)$". The significance of $\frac{dx}{dy}$ was rarely appreciated, with a significant number of candidates finding an incorrect gradient of $-\frac{1}{7}$, and confusing $\frac{dx}{dy}$ with the gradient of a normal.

Q2.

Part (a)(i) was well answered, with the main error being $\int -e^x\,dx = -2e^{2x}$. In part (a)(ii) the answer was given. Frequently candidates tried to 'fudge' their solutions to arrive at the correct result, with common error being $4\ln 2 - 2 - \frac{1}{2} = 4\ln 2 - \frac{3}{2}$, the given answer.

Part (b)(i) was again well answered by the majority of candidate, although in part (b)(ii) the step from $2x = \ln 4$ to obtain $x = \ln 2$ was often unconvincing. Some candidates answered part (c) very well and gained full credit. The main error was using 8 or –8 as the gradient of the normal, however many candidates earned at least partial marks.

Although correct answers were seen in part (d), they were not common. Many candidates earned the method mark by the substitution of $x = 0$ and some went on to find the area of the triangle. The final mark was often lost through a failure to realise that the area was positive.

Q3.

The responses to this question were very mixed, varying from the very poor to some excellent answers. Most, but not all, candidates did attempt to separate and integrate in part (a)(i) with some making an error in their integrals, and rather more omitting a constant. Such candidates could make no further progress in part (a)(ii) without a constant to find. Some candidates thought to bring e^{-1} into their answer in part (a)(i), instead of the expected conventional constant, and got themselves confused in part (a)(ii) as they could not convincingly obtain the given answer. Some candidates made the anticipated error at the exponentiation stage of $\ln y = -\cos t + c \Rightarrow y = e^{-\cos t} + e^c$. Many candidates could not justify the 50, other than saying it was the initial value.

For part (b) many candidates realised they had been given the required expression in part (a) and substituted and evaluated correctly, although some ignored the request to answer to the nearest centimetre, any greater accuracy being rather meaningless in this modelling problem. Some used degrees instead of radians. Other candidates substituted into the differential equation. In part (b)(ii), although most candidates knew in principle what to do, they could not find the second derivative correctly. Many treated y as a constant or as if it were t, rather than using implicit differentiation.

Some candidates did attempt to differentiate the given answer in part (a)(ii) twice, but usually made an error in their use of the chain rule. Some candidates did get the differentiation correct, but failed to complete their argument by demonstrating that the first derivative was zero at $t = \pi$, so confirming this point as a turning point.

Q4.

Part (a) Almost all candidates were able to find the first and second derivatives correctly, although there was an occasional arithmetic slip; some could not cope with the fraction term, others doubled 26 incorrectly.

Part (b) Those who substituted $t = 2$ into $\frac{dy}{dt}$ did not always explain that $\frac{dy}{dt} = 0$ is the condition for a stationary point. Many used the second derivative test and concluded that the point was a maximum. Some **assumed** that a stationary point occurred when $t = 2$ and went straight to the test for maximum or minimum and only scored half of the marks. A few tested $\frac{dy}{dt}$ on either side of $t = 2$ correctly, but those who only considered the gradient on one side of the stationary value scored no marks for the test.

(c) The concept of “rate of change” was not understood by many. Approximately equal numbers of candidates substituted into $t = 1$ into the expression for y, $\frac{dy}{dt}$ or $\frac{d^2y}{dt^2}$ and so only about one third of the candidates were able to score any marks on this part. Those who used $\frac{dy}{dt}$ often made careless arithmetic errors when adding three numbers.

Part (d) As in part (c), candidates did not realise which expression to use and perhaps the majority wrongly selected the second derivative. It is a general weakness that candidates do not realise that the sign of the **first** derivative indicates whether a function is increasing or decreasing at a

Q5.

This question proved to be a good source of marks for many candidates although the expansion in part (b) was quite frequently incorrect with $\left(\frac{2}{x}\right)^2$ being written as $\frac{4}{x}$ or as $\frac{2}{x^2}$. Most candidates displayed good differentiation skills in part (c) and even more realised that the answer to part (d) required evaluation of their answer to part (c) when x =

2. Even though the majority of candidates knew how to find the equation of the normal it was very common to find a mark lost in the final part of the question because candidates were unable to rearrange their answer in to the form stated in the question.

Q6.

Part (a) was well answered by the majority of candidates. Many fully correct responses were seen and, if there was an error, it was usually the addition of a function of e^x or a '+ c'.

Part (b) was answered very well with many of the candidates gaining full marks, although many then went on to try and simplify their expression, which caused a loss of marks in part (c).

Some attempts at the chain rule were seen, quite often resulting in as an $\frac{x+1}{x}$ answer. Although

many correct responses were seen, part (c) was not answered as well as the two previous parts. Most of the errors came from following through from an incorrect expression in part (b). Some candidates found the gradient of the tangent and its equation but not the normal gradient. A number of candidates left their gradients in terms of x.

Q7.

The evaluations in part (a) were done correctly by most candidates, although some made numerical errors. In part (a)(ii), the request for three significant figures was largely adhered to, although some interpreted this as three decimal places.

Candidates started part (b) well but many were confused as to how to handle the negative signs.

Many attempts to take logs of negative numbers were seen. Many got to $14\ln\left(\frac{5}{12}\right)$ or $-14\ln\left(\frac{5}{12}\right)$ and gave this as their answer. Despite this, the vast majority gave the correct answer of 12 to part (b)(ii), some just ignoring their negative sign or saying it must be positive. They apparently did not respond to a negative answer by reviewing their answer to part (b)(i). Very few candidates explained what they were doing with the negative index to obtain the expected answer. In part (c)(i), very few candidates derived the differential equation with confidence. Some made a start by attempting to differentiate the expression

for x, although, of those, many made the error of bring down the t together with $\frac{1}{14}$. Others attempted to find an expression for t and differentiate that. There was usually little progress beyond these starting points, although several candidates attempted to convince the examiner that they had found the requested result, with much abuse of the ln function seen in such attempts.

Some tried to integrate the given differential equation to recover the expression in x, and some very neat solutions were seen this way, although those who did not consider limits or a constant could make no further progress. Again, handling negative signs with logarithmic expressions caused difficulties. Most candidates scored the final mark for applying the result from part (c)(i), although some did say that $\frac{1}{14}(7)=2$.

Q8.

This question was generally done well with most candidates demonstrating at least some knowledge and ability with implicit differentiation.

For part (a), many candidates only verified that a = 1 was a possible solution and thus failed to show that it was the only positive solution. Those who did form a quadratic equation as expected usually factorised it correctly, with some giving a clear argument as to why a = 1.

In part (b), a few candidates rearranged the given equation before attempting differentiation, and some attempted to divide through by y^2, often making algebraic errors and leading to poor attempts at differentiation.

Most took the expression as given and were nearly always correct in differentiating the left hand side; they usually made a good attempt at either the product rule or chain rule, and often both, on the right hand side. Many fully correct solutions were seen. Errors varied from dropping a 5 or 2, to attaching $\frac{dy}{dx}$ to the wrong term or bringing in a spurious $\frac{dy}{dx}$ having started their solution with $\frac{dy}{dx}$ =. Some candidates did not evaluate their derivative at (1, 1) as requested.

In part (c), most candidates were able to use their answer to part (b) to find an equation of the tangent, although a few found an equation of the normal instead.

Q9.

(a) (i) The differentiation was generally well done, though some candidates found the fractional coefficient problematic. Those who wrote $6/3t^5$ were not penalised in this part but writing 6.$\frac{1}{3}$ generally led to errors later in the question. Some tried to avoid the fraction by considering $3V$ throughout the question, making errors, and suffered a heavy penalty.

(ii) Again, most applied the method correctly and here simplification of coefficients was necessary. A few failed to differentiate $6t$ or omitted it altogether.

(b) This part was poorly done with many not recognising that the rate of change was $\frac{dV}{dt}$ and substituted t = 2 into the expression for V or $\frac{d^2V}{dt^2}$. Quite a lot of candidates made arithmetic errors. A few found two values of the expression and averaged them.

(c) (i) Again, many failed to evaluate $\frac{dV}{dt}$ in order to verify that a stationary point occurred, but those who did generally obtained a value of zero. It was essential to include a relevant statement to earn both marks.

(ii) This required evaluation of the second derivative at $t = 1$ or an appropriate test. Candidates who tried to test the gradient on either side of 1 almost invariably failed, as the values used were too far away from the stationary point. A surprising number evaluated 10 – 24 + 6 to be –20 thus losing the accuracy mark. Some appeared to be guessing and drew wrong conclusions about maxima or minima after evaluating the second derivative.

Q10.

(a) Not everyone recognised that the height of the rectangle was the value of y when $x = -1$ or $x = 4$. Some who did made numerical errors. Even having found the height 4, some obtained the wrong area by taking AB as 4 or 2 (sometimes using Pythagoras' Theorem) or by finding the perimeter instead.

(b) (i) The integral was generally correct, though sometimes incorrect simplification occurred subsequently. A few integrated x^3 to $\frac{1}{4}x^4$ and a surprising number misread the integrand as $3x^2 - x^2$. There were also candidates who confused integration with differentiation or whose process was a hybrid of the two.

(ii) Almost everyone recognised that they should firstly evaluate the integral from –1 to 2, but most stopped there, instead of going on to subtract the value of the integral from the area of the rectangle. There were a lot of sign errors in the work with some adding instead of subtracting or putting the two parts the wrong way round. A few wrongly substituted in the original function. Those who chose to work with the differences of two integrals seldom completed it correctly.

(c) (i) Differentiation was done well on the whole.

(ii) Many substituted $x = 1$ into the derivative to find the gradient of the tangent and went no further. Most did not find the y coordinate of the point and so made no attempt at the equation of the tangent. A few non-linear equations were seen with a 'gradient' of $6x - 3x^2$.

(iii) Very few candidates completed this part. Many made no attempt, and those who did tended to test a few values of x or to find the second derivative, which was of no value.

Only the strongest candidates realised that $\frac{dy}{dx} < 0$ was the condition for y to be decreasing and that, after a couple of lines of algebra, the given inequality could be obtained.

(d) Although most made an attempt at the quadratic inequality, few obtained both parts of the solution. It was imperative that candidates wrote $x > 2$, $x < 0$ and not $0 > x > 2$ as many incorrectly stated.

It was disappointing to see how many candidates at this level could not solve the equation $x^2 - 2x = 0$, obtaining values such as –2, √2, 1+√2. Using the formula or completing the square sometimes led to 1 ± √1, which many candidates failed to simplify.

Q11.

Many candidates were able to differentiate y correctly but weaker candidates then had difficulty dealing with the negative index. The common error was to lose the negative sign in the power and write $16 - x^{-2} = 0$ as $16 - x^2 = 0$ which lead to the wrong values ±4. Some candidates applied the difference of two squares as the initial step in solving $16 - x^{-2} = 0$ with varying final success.

Q12.

This final question was a good source of marks for many candidates. Even the weaker candidates were able to give correct answers for parts (a) and (c) which tested standard C2 work on differentiation and integration. A significant minority of candidates started part (b)(i) by equating dy/dx to zero instead of substituting $x = 0$ to find the value of the derivative. Those who were able to answer the first two parts of part (b) correctly generally went on to find the correct coordinates of P. A significant number of candidates left their final answer as 24.3 without considering the area of the triangle OAP and carrying out the subtraction. For those candidates who attempted to find the area of the triangle some applied very lengthy methods which involved finding the length of each side of the triangle then using the cosine rule to find angle OPA then using the formula $\frac{1}{2}ab \sin C$ or involved the integration of the equations of the two

lines. To their credit a number of such candidates did avoid premature approximation to reach the correct value for the area of the triangle which could have been obtained more directly by using $\frac{1}{2} OA \times |y_P|$.

Q13.

(a) This was well answered although common incorrect answers were $3\sec^2 x$ and $\sec^2 3x$.

(b) Most candidates were able to apply the quotient rule but a large number of candidates were unable to correctly expand the numerator. A common response was $3(2x + l) - 2(3x + l) = 6x + 3 - 6x + 2$.

Q14.

(a) (i) This was reasonably well done, although some candidates integrated.

(ii) Many candidates failed to appreciate the significance of part (a)(i) and obtained complicated incorrect expressions. Those candidates who did appreciate the relevance of part (i) either obtained the correct answer or $2\ln |x^4 + 2x|$.

(b) (i) This was fairly well answered. However, many candidates lost marks because they did not realise that in any substitution question, apart from replacing functions of x with functions of u, they have to find and use $\frac{du}{dx}$.

(ii) The integration was very well answered by the majority of candidates with many accurately completing the question. Problems did arise through the use of incorrect limits. Many candidates converted their integration answers in u to terms in x and then used the original limits. This method was equally successful.

Q15.

Part (a) Most candidates answered this part correctly. A few included the 7 or thought the derivative of the first term was $7x$ and the – sign was sometimes lost.

Part (b) Many substituted $x = 1$ correctly, though it was apparent that they did not recognise this value of –10 to be the gradient of the tangent. Many correctly found y as 5 but stopped there. Again, some correct attempts at the tangent equation using $y = mx + c$ foundered and quite a large number attempted to find the equation of the normal.

Part (c) Use of the value of $\frac{\mathrm{d}y}{\mathrm{d}x}$ was the only acceptable method here. Evaluations of y at different points or finding the second derivative were common but earned no marks.

Q16.

Part (a)(i) Most candidates differentiated correctly. However a few made a slip or misread one of the terms.

Part (a)(ii) Although most managed to substitute 2 into their derivative some made numerical errors and some used y or the second derivative. Most who realised that they should equate their derivative to zero (or at least showed their intention though never inserting the = 0) then tried to factorise or use the formula (though it was clear that some did not recognise the quadratic equation as such). It was disappointing that 14 the bracket ($3x$ – 14) often produced the solution x = 14 instead of 14/3, and the solution of x = 2 did not always appear.

Part (b)(i) The integration was also completed correctly by most candidates, although the $28x$ was occasionally wrong and some 'hybrid' processes led to terms such as $\frac{20x^3}{3}$

In part (b)(ii) almost everyone attempted to substitute 3 into their integral but their problems with the ensuing fractions often took pages to resolve, and although most ended 'magically' with the required answer there were many errors en route. A few substituted into the original expression for y instead of the integrated expression.

Part (b)(iii) This part was quite well done although again there were errors in evaluating both $\frac{1}{2}$ × 21 × 3 and 56 $\frac{1}{4}$ –31 $\frac{1}{2}$. Some candidates confused length with area and merely used Pythagoras's Theorem to find the length of the hypotenuse of the triangle. Those who chose to integrate the equation of the straight line were sometimes successful but many made arithmetic errors.

Q17.

This question also proved to be a good source of marks for many candidates. The examiners expected to see some intermediate evaluation before the printed answer was quoted in part (a)(i) after the substitution of 4 for x in the given equation. Most candidates gave the correct value for the power k although the wrong values 1/2 and 1/2 were both seen in part (a)(ii). Candidates were generally able to differentiate the given expression to find the second derivative in part(a)(iii) but some believed that the sign of the x value rather than the sign of the second derivative determined the nature of the stationary point in (iv). In part (b) a significant number of candidates gave the equation of the tangent instead of the normal or used the gradient of the normal as –12. Finding the integral in part (c)(i) was well answered although, somewhat surprisingly, the integration of –7 caused as many problems as the integration of the other two terms. The final part of the question proved to be beyond many average candidates who forgot to insert the constant of integration in answering the previous part. The hint 'Hence', which should have suggested the use of the answer to (c)(ii) was not always picked up. It was not uncommon to see the equation of the curve given as a linear equation, normally the tangent at P.

Q18.

This question proved to be a good source of marks for many of the candidates. Its structure enabled candidates to progress even if they were unsuccessful with some of the parts.

Part (a)(i) Reasonably well done. Errors were with the derivative of e^{2x} Common errors were e^{2x} and $2x\ e^{2x}$ A number of candidates added a '+ c' when they were differentiating.

Part(a)(ii). Usually correct, with errors, if seen, similar to those in the previous part.

Part (b)(i) The majority of candidates answered this correctly.

Part (b)(ii) Very well answered by the majority of candidates.

Part (b)(iii) Again very well answered with many totally correct responses. The major error apart from those who used the incorrect equation $e^{2x} - 5\ e^{2x} + 6$ was the evaluation of e^{21n3} which was often seen as 6 rather than 9. Possibly the evaluation of e^{21n2} as 4 was fortuitous for some candidates.

Part (iv) Similar to part (iii) with similar errors, the major one being the evaluation of $4e^{21n3}$ as 4 × 6 = 24. This resulted in an answer of –6 and hence loss of accuracy marks.

Q19.

Many candidates lost marks on this question from careless work or failure to write answers to the correct degree of accuracy.

Part (a) Although many candidates obtained the correct method mark

$\frac{dy}{dx} = x \times \frac{1}{x} + \ln x$, a significant number who reached this stage were unable to cope with $x \times \frac{1}{x}$ and equated it to 0, giving an answer of 1n x.

Part (b) Many candidates failed to realise the connection with the previous part. A significant number of candidates just wrote down the correct answer with no method indicated.

Part (c) Many candidates scored the method mark; some for the correct substitution and many for following through their answer to (b). Some candidates lost the accuracy mark for not writing the answer in exact form but going straight from (5ln5 – 5) – (1ln1 – 1) to 4.047. Surprisingly, a number of candidates could not evaluate –5 – (–1) correctly.

Q20.

Part (a) Reasonable attempts were made by most candidates.

Part (b)(i) Many candidates answered this correctly but there were many errors such as dividing by 2 first or writing $\sin^{-1} 2x$ as $1/\sin 2x$

Part (b) (ii) This part was usually answered correctly. The usual error was the inclusion of a negative sign.

In part (c) although some candidates gave fully correct responses the majority were only able to do the dv 2 first part to get $\frac{dy}{dx} = \frac{2}{\cos y}$.

Q21.

Most candidates did well on this question, and even if some showed in part (a) that they apparently could not solve a quadratic equation, the implicit differentiation required in part (b) was generally done well.

Part (a) Most candidates substituted $x = 1$ correctly, and solved the resulting quadratic equation, usually by factorising, although some attempted a solution by non quadratic techniques, and usually made an error.

In Part (b) Many clear derivations were seen here, although candidates often dropped a mark through not equating their expression to zero during the derivation. The other relatively common error was in the product rule, where some candidates were not convincing in their use of signs and whether or not dy/dx was attached to a term.

Part (c) Most candidates successfully used the answer given to calculate the values of the derivatives, with some sign errors being made. Some used their own version of the derivative whereas some left the result in term of y, not realising they were to use the result of part (a).

Part (d) This part of the question defeated most candidates. Although most realised the derivative had to be set to zero, they could take it no further, and many who did write down $y - 6x = 0$ then couldn't see how to proceed. Some candidates actually "cross multiplied" to get$y$-6x = 2y-x, usually before abandoning the attempt. Some candidates were convinced this part of the question was to do with the second derivative and did a great deal of work for no credit. Some used the given answer of $33x^2 - 5 = 0$ and solved it for x before abandoning or doing things such as substituting into the expression for the gradient.

Q22.

(a) Most candidates knew how to differentiate, but a few made careless mistakes when differentiating $3x$ and others insisted on adding + C to their otherwise correct answer.

(b) (i) Usually sufficient working was shown to verify that the gradient at P was equal to 5.

(ii) Many seemed confused between the tangent and the normal, with many clearly trying to find the equation of the former rather than the latter. The major problem, once again, was rearranging the equation into one involving integer coefficients.

(c) It was clear that some candidates were unfamiliar with the need to consider the sign of $\mathrm{d}y/\mathrm{d}x$ to determine whether y was increasing or decreasing. It is a pity that $\frac{\mathrm{d}^2 y}{\mathrm{d}x^2}$ also had value –16 when $x = 1$, yet it was surprising to see how many felt they had to consider the sign of the second derivative rather than the first.

Q23.

(a) Once candidates realised that the lengths in centimetres of the sides of the box were $24 - 2x$, $9 - 2x$ and x, they were usually able to multiply out to give the printed answer for the volume. A large number of weaker candidates found this part of the question very difficult and made little or no effort to obtain the printed answer. (a)(i) The expression for $\mathrm{d}V/\mathrm{d}x$ was usually correct.

(ii) This part was answered too casually by many candidates who simply divided their previous answer by 12 and wrote "= 0" to comply with the printed answer. It was necessary to see that the candidate realised that it is necessary for $\frac{\mathrm{d}V}{\mathrm{d}x} = 0$ for stationary points to occur, before dividing their expression for $\mathrm{d}V/\mathrm{d}x$ by 12.

(iii) Most candidates scored the marks for solving the quadratic equation, although aq few who chose to use the formula, rather than factorising, made slips in their arithmetic.

(c) It was pleasing to see that most candidates were able to find a second derivative. However, this was often given as $2x - 11$, since many had previously divided $\mathrm{d}V/\mathrm{d}x$ by 12 and were using this modified expression. Some candidates did not appear to understand the second derivative test for maxima and minima; they wrote things such as " $\frac{\mathrm{d}^2 V}{\mathrm{d}x^2} = 24x - 132 = 0$ when $x = 5.5$ and since this is positive the stationary point is a minimum".

Q24.

Parts (a)(i) and (a)(ii) were generally well answered although it was surprising to find the derivative of x seemingly causing as many problems as the derivative of $2/x$. Although the method for finding the equation of a line was generally well known, a significant minority of candidates found the equation of the tangent rather than the equation of the normal.

Q25.

Use of the binomial expansion and use of the expansion of $(2 + x)(2 + x)^2$ were equally popular in part (a)(i) and most obtained the correct answer. There was however a minority of candidates who did not include the powers of 2 in the middle terms when applying the binomial expansion. It was disappointing to see so many candidates producing masses of algebra rather than replacing x by $(-x)$ to obtain the answer to part (a)(ii). There was some evidence of 'faking' to reach the printed answer in part (b), which quite often resulted in marks being lost as earlier correct work was changed. The vast majority realised that dy/dx was required in part (c), although some later explanations could have been more precise.

Q26.

This question was answered very well by many candidates, and evidently the basic processes of differentiation and integration had been well rehearsed.

Part (a)(i) Most candidates substituted $x = 3$ into the given cubic equation and found $p(x)$ to be zero. Some, however, lost a mark as they did not indicate that $p(3) = 0$ implied that $x - 3$ was a factor.

Part (a)(ii) The coordinates (3, 0) were often seen but some forgot to give the y coordinate

Part (b)(i) Almost all candidates differentiated correctly, which was pleasing.

Part (b)(ii) Most candidates realised the need to equate dy/dx to zero. However not all were able to solve the resulting equation. Some of those who did solve the quadratic offered both solutions, 1 **and** 7/3. Many found the second derivative in order to ascertain which point was the minimum point M. This was unnecessary as the coordinates of A (1, 0) were clear from the diagram.

Part (c) The second derivative was almost always found correctly followed by the substitution of $x = 1$.

Part (d)(i) The integration was carried out very well. Occasionally $\frac{3x^4}{4}$ was seen but this was rare.

Part (d)(ii) Most candidates used the correct limits of 0 and 1. However many arithmetic errors occurred.

When trying to combine the fractions so that the correct value of the integral $-\frac{11}{12}$ was not seen very often. Of those who did integrate correctly, many still lost the last mark for not stating that the area of the shaded region was 11/12 and the negative value of the integral arose because the region was below the x-axis. It was not acceptable to simply write Area = $-\frac{11}{12}=\frac{11}{12}$.

Q27.

This question proved to be a good source of marks for many candidates. In part (a)(i) most candidates quoted the correct value for the power p. Although many candidates produced correct answers for the integration, all the usual errors, as illustrated by $\frac{3}{2}x^{1.5}, x^{1.5}, \frac{x^{-0.5}}{-0.5}$, and $\frac{\sqrt{x}^2}{2}$, were seen. Most candidates stated and used the correct limits in part (a)(iii) but evaluation was sometimes poor. Many of the candidates found a correct equation for the tangent but it was surprising to find a significant minority of these candidates either making no attempt to find the area of triangle AOB or giving the area as a negative value. For full marks in part (c) the examiners were expecting to see the word ‘translation’. Some candidates wrote ‘tr’ but this was not considered to be clear enough to distinguish between ‘transformation’ and ‘translation’. Common wrong answers involved ‘stretch’ and vectors in the wrong direction. In part (d) there was a significant minority of candidates starting their solution with the incorrect statement $\sqrt{x-1}=\sqrt{x}-\sqrt{1}=\sqrt{x}-1$ but in general the trapezium rule was well understood and many candidates were able to give the final answer to the correct required degree of accuracy in this numerical integration question.

Q28.

Most candidates found one correct term but only a minority found both. The common incorrect answers are illustrated by ‘$x^8 - x^{-3}$ and ‘$x^5 - x^3$’Most candidates were able to differentiate their expression for f(x), but the method required to show that f is an increasing function was not well known. Many did not attempt it and many others tried to justify it by considering the relative size of f ′ (x) for two particular values of x. Others incorrectly stated that ‘f ′(2) > f (2) so f is increasing’ or used the second derivative. Those who considered the sign of f ′(x) for the given domain, $x > 0$, usually gained both marks in a line of working. Those candidates who realised that the gradient of the tangent was given by f ′(1) usually obtained a correct follow through answer for the gradient of the normal. It is worth noting that many candidates went on to find the equation of the normal. Although, of course, this further work was not penalised, it may have used up valuable time for some of the candidates.

Q29.

Part (a) was well answered, although some candidates integrated. There were many fully correct answers to part (b), but a lot of candidates chose the incorrect functions to equal u and dv/dx with the obvious ensuing problem.

Part (c) was poorly answered. Candidates must realise that, in any substitution question, apart from replacing functions of x with functions of u, they must also find du/dx. A large number of candidates attempted to integrate an expression containing both x's and u's.

Q30.

This question was also answered well by virtually all candidates, who showed that they knew the techniques involved in answering it. In part (c) the majority of students integrated to find the position vector but a common error was to forget the + c term and thus to ignore the initial position vector. Some found the + c incorrectly assuming that **c** was zero without completing the necessary calculation.

Q31.

Most candidates showed that they knew the techniques involved in answering this question. Candidates generally answered part (a) correctly. Part (b)(i) was answered well, but in part (b)(ii) some candidates forgot to find the magnitude of **F**, giving instead just the vector **F**, – 40**i** + 30**j**. Most candidates answered part (c) correctly. In part (d) the majority of candidates integrated to find the position vector, but a common error was to forget the +**c** term and thus to ignore the initial position vector. Some found the + **c** incorrectly, assuming that **c** was the initial position vector.

Q32.

This question was answered well; a few candidates had difficulty in the differentiation of $3\cos 4t$.

Q33.

Virtually all candidates answered this question well, with only a few forgetting to find the magnitude of the force in part (c).

Q34.

Part (a) was usually answered correctly, although some obtained $+6\cos 3t$, rather than $-6\cos 3t$. In part (b), the common error was the omission of $+c$, while others added $+c$ and automatically assumed that $c = 0$.

Q35.

This question was generally answered well. The only common error was in part (b)(ii), where a number of candidates found **F** to be 36**i** + 6**j**, but did not find its magnitude.